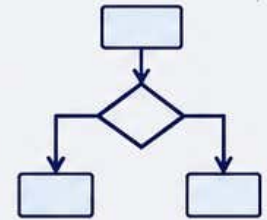


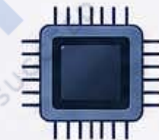
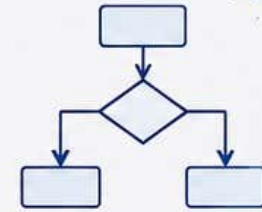
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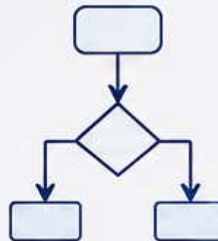
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Introduction and Data Preprocessing

Data mining is the process of discovering useful patterns, trends and knowledge from large amounts of data. Data preprocessing converts raw data into a clean, consistent and high-quality format suitable for mining.

1. Introduction to Data Mining

• Definition :

Data mining is the process of extracting hidden, valid, novel, useful and understandable patterns from large datasets.

• Goal :

- Discover meaningful patterns and relationships.
- Make predictions and support decision making.
- Extract knowledge from historical data.

- Data Mining is the step in Knowledge Discovery in Databases (KDD) process.



• Applications :

- Business : Market analysis, customer segmentation, sales prediction.
- Banking : Fraud detection, credit scoring, risk analysis.
- Healthcare : Disease prediction, patient monitoring.
- Education : Performance prediction, student behavior analysis.
- Others : Web usage mining, social media analysis, etc.

• Data Mining Tasks :

- | | |
|---------------------------|-----------------------------|
| ⇒ Classification | ⇒ Sequential Pattern Mining |
| ⇒ Regression | ⇒ Outlier Detection |
| ⇒ Clustering | ⇒ Summarization |
| ⇒ Association Rule Mining | ⇒ Prediction |

2. Data Preprocessing

• Definition :

Data preprocessing is the process of converting raw data into a clean, consistent and useful format for analysis and data mining.

• Importance :

- Improves data quality.
- Increases accuracy and efficiency of mining algorithms.
- Reduces noise, inconsistency and missing values.

• Steps in Data Preprocessing :

- ① Data Cleaning → Handle missing values, remove noise and correct inconsistencies.
- ② Data Integration → Combine data from multiple sources into a consistent format.
- ③ Data Transformation → Normalize or aggregate data into suitable forms for mining.
- ④ Data Reduction → Reduce the size of data while preserving important information.
- ⑤ Data Discretization → Convert continuous data into categorical (discrete) values.

Data Cleaning Techniques :

Problem	Technique
Missing Values	Ignore tuple, fill manually, use mean/median/mode, prediction methods.
Noisy Data	Smoothing, binning, regression, clustering.
Inconsistent Data	Data validation, standardization.
Outliers	Detect and remove or replace using thresholds.

In Short : Data mining discovers knowledge from data, while data preprocessing prepares high-quality data, which is a crucial first step for successful data mining.

Need and Applications of Data Mining

Data mining is the process of discovering useful patterns, trends and knowledge from large volumes of data to support decision making and improve performance.

1. Need of Data Mining

- **Explosion of Data** : Huge amount of data is generated every day from various sources.
- **Hidden Patterns** : Useful information is hidden in large data and not visible through manual analysis.
- **Better Decision Making** : Helps organizations make data-driven decisions with more accuracy and confidence.
- **Time and Cost Saving** : Automates analysis of large datasets, reducing manual effort, time and cost.
- **Improve Efficiency** : Optimizes business processes and resource utilization.
- **Competitive Advantage** : Helps organizations understand customers better and stay ahead of competitors.
- **Risk Reduction** : Identifies potential risks and prevents problems before they occur.
- **Forecasting and Prediction** : Predicts future trends and behaviors based on historical data.

In Short : Data mining is needed to convert large, complex and raw data into meaningful knowledge for better decisions, improved efficiency, reduced risks and gaining competitive advantage.



2. Applications of Data Mining

Data mining is used in almost every domain. Some major applications are :



Business / Retail : Market basket analysis, customer segmentation, sales forecasting, product recommendation.



Banking & Finance : Fraud detection, credit scoring, risk analysis, customer profiling, loan approval prediction.



Healthcare : Disease prediction, patient monitoring, treatment analysis, healthcare management.



Education : Student performance analysis, dropout prediction, course recommendation, learning analytics.



Telecommunications : Churn prediction, fraud detection, network optimization, customer satisfaction analysis.



Manufacturing : Quality control, fault detection, predictive maintenance, demand forecasting.



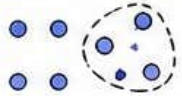
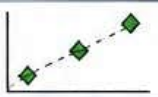
Government & Security : Crime analysis, threat detection, resource allocation, policy planning.



Others : Web usage mining, social media analysis, weather prediction, agriculture, insurance, etc.

Data Mining Functionalities

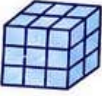
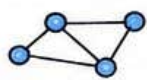
Data mining functionalities are the major data mining tasks that discover useful patterns and knowledge from large datasets.

Sr. No.	Functionality	Description	Example
①	 Characterization (Summarization)	Generalize the general properties of the data in a target class.	<u>Example</u> : Find the common profile of high-value customers.
②	 Discrimination (Class Comparison)	Compare the characteristics of a target class with other classes.	<u>Example</u> : Compare the buying behavior of loyal customers and non-loyal customers.
③	 Association Rule Mining	Discover relationships or associations among items in large datasets.	<u>Example</u> : Customers who buy bread also tend to buy butter.
④	 Sequential Pattern Mining	Discover frequent sequences or ordered patterns over time.	<u>Example</u> : A customer first buys mobiles, then buys accessories, then buys insurance.
⑤	 Clustering	Divide data into groups (clusters) such that data within a cluster are more similar to each other than to those in other clusters.	<u>Example</u> : Group customers into clusters based on their purchasing behavior.
⑥	 Regression	Predict continuous or real-valued attributes.	<u>Example</u> : Predict the future sales amount based on past sales data.
⑦	 Classification	Predict categorical (discrete) class labels.	<u>Example</u> : Predict whether a customer will buy a product or not.
⑧	 Anomaly (Outlier) Detection	Identify rare items, events or observations that do not conform to the general behavior.	<u>Example</u> : Detect unusual credit card transactions (fraud detection).
⑨	 Evolution / Trend Analysis	Analyze changes and trends in data over time.	<u>Example</u> : Analyze monthly customer growth or sales trend over years.

In Short: These functionalities help in extracting valuable knowledge from data and support better decision making in various real-world applications.

Types Of Data in Data Mining

Data mining is applied on different types of data. The nature and structure of data determine the mining techniques and the kind of patterns that can be discovered.

Sr. No.	Type of Data	Description	Examples	Characteristics
1.	 Relational Data	Data stored in tables (rows and columns) with a fixed schema.	- Customer table, Sales table, Product table in RDBMS - Data in SQL databases	- Structured - Easy to store and query - Most common in data mining
2.	 Multidimensional Data	Data organized in multidimensional structures (cubes) for analysis across multiple dimensions.	- Sales by time, region, product - Data warehouse / OLAP cube	- Structured - Fast aggregation and summarization - Useful for decision support
3.	 Transactional Data	Data that records transactions or events.	- Bank transactions - Purchase in shopping carts - Web clickstream data	- Time-stamped - Large volume - Used for association, sequential pattern mining
4.	 Text Data	Unstructured data in the form of natural language text.	- Emails, documents, articles - Customer reviews, social media posts	- Unstructured - Requires NLP techniques - Useful for text mining, sentiment analysis
5.	 Image / Video Data	Data in the form of images or videos.	- Medical images (MRI, X-ray) - CCTV videos, satellite images - Product images	- Unstructured - Requires image / video processing - Used in pattern recognition
6.	 Audio / Speech Data	Data in the form of sound or speech.	- Voice recordings - Call center conversations - Speech commands	- Unstructured - Requires signal processing - Used in speech recognition
7.	 Graph / Network Data	Data that represents entities as nodes and relationships as edges.	- Social networks - Web link data - Communication networks	- Highly connected - Complex relationships - Used in social network analysis, link prediction
8.	 Time-Series Data	Data points collected or recorded over time intervals.	- Stock prices over time - Sensor readings, weather data - Sales over months / years	- Time ordered - Shows trends and seasonality - Used in trend analysis, forecasting

In Short : Data in data mining can be structured (relational, multidimensional), semi-structured (transactional, text) or unstructured (image, audio, graph, time-series). Understanding the type of data is essential to choose the right mining techniques.

Data Preprocessing : Data Cleaning , Data Integration

Data preprocessing converts raw data into a clean, consistent and integrated form suitable for accurate analysis and data mining.

1. Data Cleaning

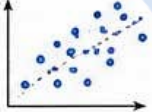
→ Definition :

Data cleaning is the process of identifying and correcting or removing inaccurate, incomplete, inconsistent, noisy and duplicate data.

→ Need :

- Improves data quality and reliability.
- Increases accuracy of analysis and models.
- Removes noise and inconsistencies in data.

→ Common Problems in Data :

Missing Values	Noisy Data	Inconsistent Data	Duplicate Data	Outliers															
<table border="1"><tr><td>Age</td></tr><tr><td>25</td></tr><tr><td>—</td></tr><tr><td>30</td></tr><tr><td>28</td></tr></table>	Age	25	—	30	28		<table border="1"><tr><td>M/F</td></tr><tr><td>Male</td></tr><tr><td>m</td></tr><tr><td>FEMALE</td></tr></table>	M/F	Male	m	FEMALE	<table border="1"><tr><th>ID</th><th>Name</th></tr><tr><td>101</td><td>Amit</td></tr><tr><td>101</td><td>Amit</td></tr></table>	ID	Name	101	Amit	101	Amit	
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28																			
M/F																			
Male																			
m																			
FEMALE																			
ID	Name																		
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101	Amit																		
<u>Example :</u> Age is missing for a record.	<u>Example :</u> Incorrect or random data.	<u>Example :</u> Same information in different ways.	<u>Example :</u> Same record appears again.	<u>Example :</u> Values far from normal range.															

→ Data Cleaning Tasks :

- Handling Missing Values → Deletion, Mean / Median / Mode, Interpolation.
- Smoothing Noisy Data → Binning, Regression, Clustering, Filters.
- Resolving Inconsistencies → Standardization, Data type conversion.
- Removing Duplicates → Duplicate detection and removal.
- Detecting and Handling Outliers → Using statistical methods or domain knowledge.

In Short : Data cleaning improves the quality of data by removing errors, inconsistencies and noise, which leads to better mining results.

2. Data Integration

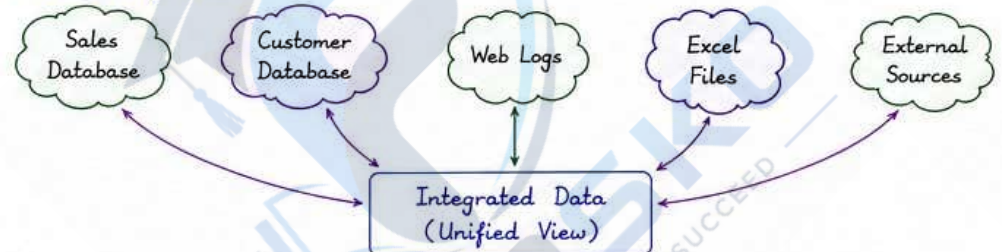
→ Definition :

Data integration combines data from multiple sources into a consistent and unified view.

→ Need :

- Provides a comprehensive view of data.
- Helps in better decision making.
- Eliminates redundancy and inconsistency across sources.

→ Sources of Data :



→ Data Integration Tasks :

- Schema Integration → Matching attributes, resolving schema conflicts.
- Entity Resolution → Identifying the same real-world entity.
- Redundancy Elimination → Removing duplicate data across sources.
- Consistency Check → Ensuring data follows business rules.
- Data Transformation → Converting data into a common format.

Example :

Sales DB			Customer DB			Web Data		→	Integrated Data		
Cust_ID	Name	City	ID	Name	City	Customer	Location		Cust_ID	Name	City
1	Rahul	Pune	1	Rahul	Pune	Neha	Mumbai		1	Rahul	Pune
2	Neha	Mumbai	3	Amit	Delhi	Amit	Delhi		2	Neha	Mumbai
									3	Amit	Delhi

In Short : Data integration combines data from different sources and provides a unified and consistent dataset for analysis and mining.

Data Preprocessing : Data Transformation, Data Reduction

Data transformation converts data into a suitable form for mining. Data reduction reduces the size of data while preserving important information.

3. Data Transformation

Definition : Data transformation is the process of converting data from one format or structure into another appropriate format for data mining.

Need :

- Improves the quality and consistency of data.
- Converts data into a form suitable for analysis.
- Handles heterogeneity in data.
- Helps in improving the efficiency of mining algorithms.

Common Transformation Techniques :

- ① Smoothing
(Removing noise)



- ② Normalization
(Scaling data)

Before (Age)					After (0 to 1 scale)				
10	25	40	70	90	0.00	0.19	0.37	0.75	1.00

- ③ Aggregation
(Summarization)

Daily Sales		Weekly Sales	
Day	Sales (₹)	Week	Total Sales (₹)
1	1200	Week 1	4850
2	1100		
3	1300		
4	1250		

- ④ Attribute / Feature Construction
(Creating new attributes)

Existing Attributes		New Attribute	
Length (cm)	Width (cm)	Area (cm ²)	
10	5	50	

- ⑤ Generalization
(Concept hierarchy)

Specific Data		Generalized Data	
Pune, Mumbai, Nagpur, Nashik, Aurangabad		Maharashtra	

- ⑥ Discretization
(Continuous to discrete)

Continuous Age					Discretized Age			
22	27	35	45	52	20-30	31-40	41-50	>50

In Short : Data transformation standardizes and converts data into the required format to make mining more effective and efficient.

4. Data Reduction

Definition :

Data reduction is the process of reducing the volume of data while minimizing the loss of important information.

Need :

- Reduces storage space requirement.
- Speeds up data mining algorithms.
- Eliminates irrelevant or redundant data.
- Makes data easier to visualize and analyze.

Common Data Reduction Techniques :

- ① Dimensionality Reduction

Reduces the number of attributes.

Example :

From 100 attributes → select 10 most important attributes.

- ② Numerosity Reduction

Reduces the number of records.

Example :

From 1,00,000 customer records → select 10,000 representative records.

- Sampling
- Histograms
- Clustering
- Regression
- Data Cube Aggregation

- ③ Data Compression

Uses compressed representation of data.

Example :

Compress text data using ZIP, images using JPEG.

- ④ Discretization & Binning

Converts continuous data into intervals (bins).

Example :

Continuous Income (₹)				Binned Income		
12,500	18,200	32,000	48,500	Low (<20K)	Medium (20K - 40K)	High (>40K)
12,500	18,200	25,000	32,000			48,500

- ⑤ Data Cube Aggregation

Aggregates data in data cubes.

Example :

Daily sales data → aggregated to monthly sales data.

In Short : Data reduction reduces the size of data by removing or summarizing irrelevant data while preserving important information, leading to efficient storage and faster mining.

Introduction to Data Warehouse

A Data Warehouse (DW) is a centralized repository that stores integrated, subject-oriented, time-variant and non-volatile data to support management decision making.

1. What is a Data Warehouse ?

Data Warehouse is a large collection of historical data that is extracted from multiple heterogeneous sources, integrated and organized for efficient querying, analysis and reporting to support decision making.

2. Characteristics of Data Warehouse

- **Subject-Oriented** : Organized around major subjects like customer, product, sales, etc.
- **Integrated** : Data from multiple sources is integrated into a consistent format.
- **Time-Variant** : Data is associated with a time element (e.g., daily, monthly, yearly).
- **Non-Volatile** : Once data is loaded, it is not updated or deleted, only appended.

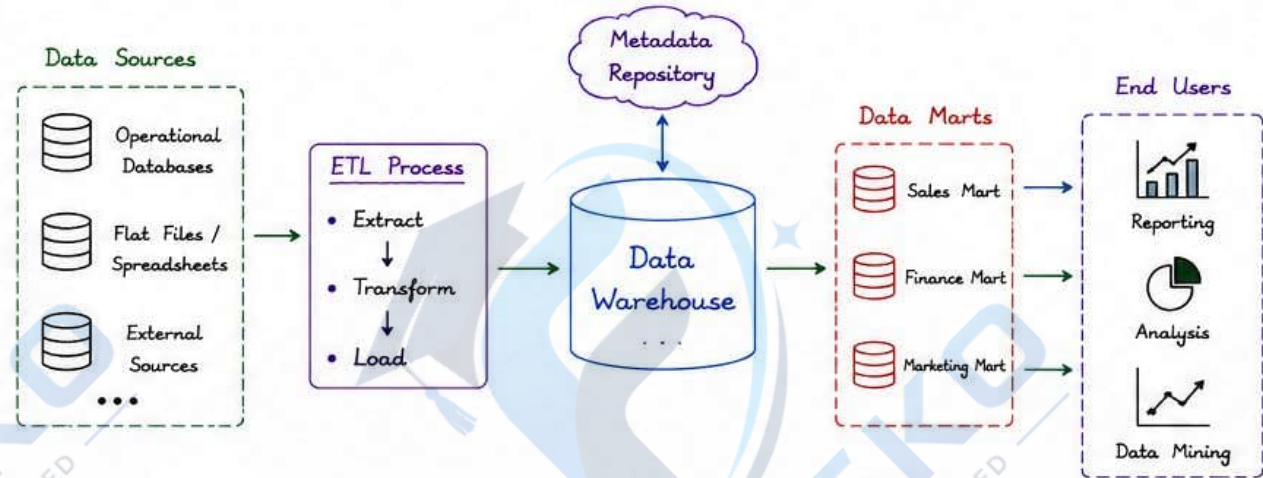
3. Need for Data Warehouse

- To support strategic decision making.
- To provide historical and summarized data.
- To integrate data from different sources.
- To improve data quality and consistency.
- To enable fast and efficient querying and reporting.

In Short : A Data Warehouse provides a clean, consistent and integrated view of data from different sources to help organizations analyze past performance and make better future decisions.



4. Architecture of Data Warehouse



5. Components of Data Warehouse

- **Data Sources** : Operational DBs, files, external systems, web data, etc.
- **ETL (Extract, Transform, Load)** : Extract data → Clean & transform → Load into DW.
- **Data Warehouse** : Central repository of integrated data.
- **Metadata Repository** : Stores information about data (data about data).
- **Data Marts** : Department-wise subset of data warehouse.
- **Front-End Tools** : Tools for reporting, OLAP, analysis, and data mining.

6. Benefits of Data Warehouse

- Improved decision making
- Better data quality and consistency
- Fast and easy access to data
- Historical data for trend analysis
- Reduced load on operational systems
- Supports OLAP and Data Mining.

Example : A company collects data from multiple sources like sales, billing, customer service, and stores it in a data warehouse. → Managers can analyze sales trends, customer behavior, and product performance for better decisions.

Difference between Data Mining and Data Warehouse

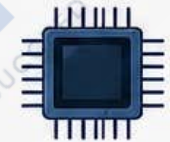
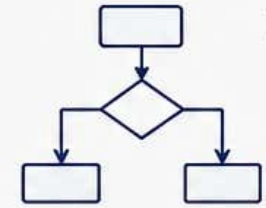
Data Warehouse stores and manages data, whereas Data Mining analyzes data to discover useful patterns and knowledge.

Sr. No.	Data Warehouse	Basis of Difference	Data Mining
1.	Data Warehouse is a centralized repository that stores integrated data from multiple sources.	Definition	Data Mining is the process of analyzing data to extract useful patterns and knowledge.
2.	It is used for storing, managing and retrieving data for reporting and decision making.	Primary Purpose	It is used for discovering hidden patterns, trends and relationships in data.
3.	It mainly supports decision support systems (DSS), reporting, OLAP and business intelligence.	Main Use	It mainly supports knowledge discovery, pattern recognition and predictive analysis.
4.	It stores large volume of historical and current data.	Data Nature	It works on current as well as historical data stored in data warehouse.
5.	Data Warehouse is a data storage system.	Function	Data Mining is a data analysis process.
6.	It organizes data (Extract, Transform, Load) and provides a unified view.	Role in Organization	It analyzes data using various algorithms and techniques to find useful information.
7.	It is subject-oriented, integrated, time-variant and non-volatile.	Characteristics	It may work on a subset of data and results in patterns, rules or models.
8.	It is a prerequisite for data mining. (Provides data)	Relationship	It is performed on data stored in data warehouse or other data sources.
9.	Data Warehouse handles structured data (mainly relational).	Type of Data	Data Mining can handle structured, semi-structured and unstructured data.
10.	It is mainly handled by database administrators and data warehouse specialists.	Performed By	It is performed by data analysts, data scientists and domain experts.
11.	Examples: Teradata, Oracle DW, Microsoft SQL Server DW, etc.	Examples	Examples: Classification, Clustering, Association Rule Mining, Prediction, Outlier Detection, etc.
12.	It does not discover new knowledge.	Output	It discovers new knowledge, patterns, rules, models and predictions.

In Short : Data Warehouse is about storing the right data.

Data Mining is about finding the right information from the data.

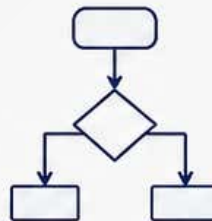
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Association Rules and Classification

Association Rules and Classification are two important data mining tasks. Association Rules help find relationships among items, while Classification predicts the class or category of data.

1. Frequent Patterns and Association Rules

→ **Association Rule**: An implication of the form $X \Rightarrow Y$, where X and Y are disjoint itemsets.

• Example: $\{\text{Bread, Butter}\} \Rightarrow \{\text{Milk}\}$

→ **Support and Confidence**

- **Support(X)**: Proportion of transactions containing X .
- **Confidence($X \Rightarrow Y$)**: Proportion of transactions containing X that also contain Y .

→ **Example Dataset (Market Basket Transactions)**

TID	Items Purchased
T1	Bread, Milk, Butter
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Coke
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Butter, Diaper

Step 1: Frequent 1-itemsets

Item	Support Count	Support
Bread	4	4/5 = 0.80
Milk	4	4/5 = 0.80
Butter	2	2/5 = 0.40
Diaper	4	4/5 = 0.80
Beer	3	3/5 = 0.60
Eggs	1	1/5 = 0.20
Coke	1	1/5 = 0.20

Step 2: Frequent 2-itemsets (min support = 0.4)

Itemset	Support Count	Support	Frequent?
{Bread, Milk}	3	3/5 = 0.60	Yes
{Bread, Diaper}	2	2/5 = 0.40	Yes
{Milk, Diaper}	3	3/5 = 0.60	Yes
{Bread, Butter}	2	2/5 = 0.40	Yes
{Diaper, Beer}	2	2/5 = 0.40	Yes
{Milk, Butter}	1	1/5 = 0.20	No

Step 3: Association Rules (min confidence = 0.6)

Rule ($X \Rightarrow Y$)	Support	Confidence	Result
$\{\text{Bread}\} \Rightarrow \{\text{Milk}\}$	3/5 (0.60)	3/4 (0.75)	Strong
$\{\text{Milk}\} \Rightarrow \{\text{Bread}\}$	3/5 (0.60)	3/4 (0.75)	Strong
$\{\text{Bread}\} \Rightarrow \{\text{Diaper}\}$	2/5 (0.40)	2/4 (0.50)	Weak
$\{\text{Diaper}\} \Rightarrow \{\text{Bread}\}$	2/5 (0.40)	2/4 (0.50)	Weak
$\{\text{Milk}\} \Rightarrow \{\text{Diaper}\}$	3/5 (0.60)	3/4 (0.75)	Strong
$\{\text{Diaper}\} \Rightarrow \{\text{Milk}\}$	3/5 (0.60)	3/4 (0.75)	Strong

In Short: Frequent patterns are itemsets that appear frequently in the database. Association rules are generated from these patterns to find interesting relationships between items.

2. Classification

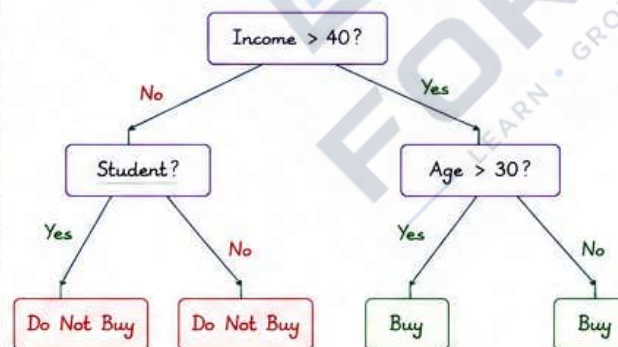
→ Classification is the process of learning a model (from labeled data) to predict the class label of new or unseen data.

→ **Example**: Classify a customer as "Buy" or "Do Not Buy" based on features.

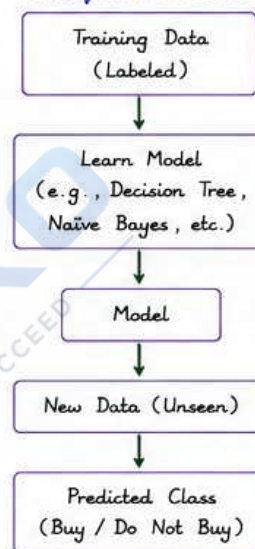
Training Data (Labeled)

ID	Age	Income (Thousand \$)	Student	Buys Laptop
1	25	35	No	Do Not Buy
2	30	50	No	Buy
3	22	20	Yes	Do Not Buy
4	35	70	No	Buy
5	28	45	Yes	Buy
6	40	80	No	Buy
7	24	25	Yes	Do Not Buy
8	33	60	No	Buy

Example using Decision Tree (simplified)



Classification Process



Buy → Positive class
Do Not Buy → Negative class

In Short: Classification builds a model from labeled data to predict the class of new records. It is widely used in decision making, medical diagnosis, email spam detection, customer churn prediction, etc.

Basic Concepts of Association Rule Mining

Association Rule Mining is used to discover interesting relationships and co-occurrence patterns among items in large transactional databases.

1. Transactions and Itemsets

→ A transaction is a set of items bought or occurring together.

→ An itemset is a set of one or more items.

Example :

TID	Items Purchased
T1	{Milk, Bread, Butter}
T2	{Bread, Diaper, Beer, Eggs}
T3	{Milk, Diaper, Beer, Coke}
T4	{Bread, Milk, Diaper, Beer}
T5	{Bread, Milk, Butter, Diaper}

Items = {Milk, Bread, Butter, Diaper, Beer, Eggs, Coke}

Example Itemsets :

1-itemset : {Milk}, {Bread}, ...

2-itemset : {Milk, Bread}, ...

3-itemset : {Bread, Milk, Butter}, ...

2. Support

→ Support of an itemset is the proportion of transactions containing that itemset.

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

Example :

In the above dataset (5 transactions)

$\text{Support}(\{\text{Milk, Bread}\}) = 3/5 = 0.60$

(Appears in T1, T4, T5)

Minimum Support (min_sup):

A threshold specified by the user.

Only itemsets with support $\geq$ min_sup are considered frequent.

3. Confidence

→ Confidence of a rule $X \Rightarrow Y$ is the conditional probability that a transaction containing X also contains Y.

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Example :

$\text{Confidence}(\{\text{Milk, Bread}\} \Rightarrow \{\text{Butter}\})$

$= \frac{\text{Support}(\{\text{Milk, Bread, Butter}\})}{\text{Support}(\{\text{Milk, Bread}\})}$

$= \frac{2}{3} = 0.67$ (Appears in T1, T5)

Minimum Confidence (min_conf):

A threshold specified by the user.

Only rules with confidence $\geq$ min_conf are considered strong.

4. Association Rule

→ An association rule is an implication of the form $X \Rightarrow Y$, where X and Y are disjoint itemsets.



Example :

$\{\text{Milk, Bread}\} \Rightarrow \{\text{Butter}\}$

(If a customer buys Milk and Bread, then he is likely to buy Butter.)

5. Lift (Interestingness Measure)

→ Lift measures how much more often X and Y occur together than if they were independent.

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}$$

Example :

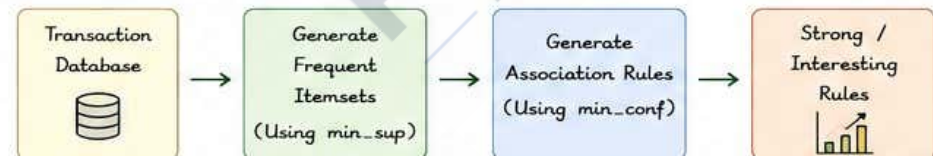
$\text{Support}(\{\text{Butter}\}) = 3/5 = 0.60$

$\text{Lift}(\{\text{Milk, Bread}\} \Rightarrow \{\text{Butter}\})$

$= 0.67 / 0.60 = 1.11$

- Lift > 1 → Positive correlation (more likely together)
- Lift = 1 → Independent
- Lift < 1 → Negative correlation (less likely together)

6. Process of Association Rule Mining



In Short : • Find itemsets that occur frequently (high support).

• From these, generate rules with high confidence.

• Evaluate interestingness using measures like lift.

• These rules help in decision making, recommendation, cross-selling, market basket analysis, etc.

Applications :

• Market Basket Analysis

• Cross-selling and Up-selling

• Recommendation Systems

• Web Usage Mining

• Inventory Management, etc.

Classification and Prediction

Classification and Prediction is a data mining technique that uses known (labeled) data to build a model which can be used to predict the class or value of unknown (unseen) data.

1. Classification

- It is the process of learning a model from labeled (training) data to predict the class label of new data.
- The predicted value is categorical (discrete).

Example : Classify an email as "Spam" or "Not Spam".

2. Prediction (Regression)

- It is the process of learning a model from labeled data to predict a continuous (numeric) value.
- The predicted value is numerical.

Example : Predict the price of a house, sales amount, temperature, etc.

3. Process of Classification and Prediction



4. Example of Classification

Training Data (Labeled)

ID	Outlook	Temperature	Humidity	Windy	Play Tennis
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
...	...	...	...	...	...

Classification Model (Decision Tree)



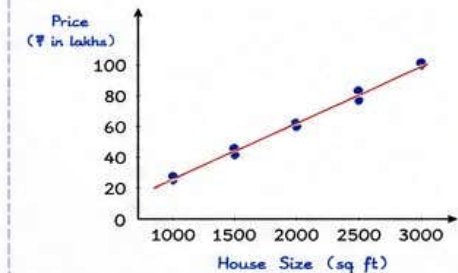
Here, the model learns patterns from the training data and can classify new records (e.g., whether to play tennis or not).

5. Example of Prediction (Regression)

Training Data (Labeled)

House Size (sq ft)	No. of Rooms	Age of House (years)	Price (₹ in lakhs)
1000	2	10	25
1500	3	15	38
2000	3	20	55
2500	4	15	63
3000	5	10	80
...	...	...	...

Regression Model (Linear Regression)



The model learns the relationship between features (size, rooms, age) and price, and predicts the price of a new house.

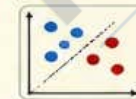
6. Important Terms

- Labeled Data** : Data with known output (class/target).
- Training Data** : Labeled data used to build the model.
- Test Data** : Unseen data used to evaluate the model.
- Model** : Representation of patterns learned from training data.
- Class Label** : The target category (e.g., Yes/No, Spam/Not Spam).
- Feature / Attribute** : Input variables used for prediction.
- Accuracy** : Measure of correctness of the model.

7. Techniques Used

Classification Techniques

- Decision Trees
- Naïve Bayes
- k-Nearest Neighbors (k-NN)
- Support Vector Machines (SVM)
- Neural Networks, etc.



Prediction (Regression) Techniques

- Linear Regression
- Multiple Regression
- Regression Trees
- Support Vector Regression
- Neural Networks, etc.



In Short : Classification predicts categorical labels (classes), while prediction estimates continuous values. Both use historical data to build models that can be applied to new, unseen data.

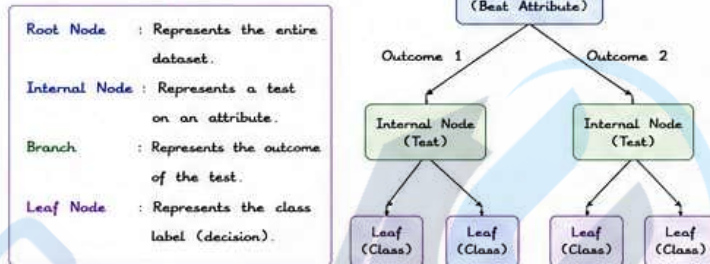
Decision Tree Classification

A Decision Tree is a flowchart like structure in which each internal node represents a test on an attribute (feature), each branch represents the outcome of the test and each leaf node represents a class label.

1. What is Decision Tree?

- It is a supervised learning technique used for classification.
- It recursively partitions the data into subsets based on feature tests.
- The result is a tree structure that is easy to understand and can be used for prediction.

2. Structure of Decision Tree



3. How Decision Tree is Built?

1. Select the best attribute that best splits the data. (using measures like Information Gain / Gini Index)
2. Split the dataset into subsets based on the values of that attribute.
3. Repeat the process recursively for each subset.
4. Stop when all records in a node belong to the same class or no more attributes are left. The final leaf node assigns the class label.

4. Example Dataset

ID	Outlook	Temperature	Humidity	Windy	Play Tennis
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes

5. Attribute Selection Measures

• Information Gain (ID3, C4.5)

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \times \text{Entropy}(S_v)$$

Where,

S : set of training examples

A : attribute

S_v : subset of S for which attribute A has value v

|S| : number of examples in S

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

c : number of classes, p_i : proportion of class i in S

• Gini Index (CART)

$$\text{Gini}(S) = 1 - \sum_{i=1}^c (p_i)^2$$

Attribute with highest Information Gain or lowest Gini Index is chosen for split.

6. Building the Decision Tree

Using the above dataset (Play Tennis)

Step 1 : Choose the best attribute for Root Node

Information Gain is calculated for all attributes.

Attribute	Information Gain
Outlook	0.246
Temperature	0.029
Humidity	0.151
Windy	0.048

Outlook has the highest Information Gain
→ Choose as Root Node.

Step 2 : Split the data based on Outlook

We get three subsets: Sunny, Overcast, Rainy.

Step 3 : Repeat the process for each subset.
Continue until all leaf nodes contain records of only one class.

7. Final Decision Tree for Play Tennis



8. How to Classify New Instance?

- Start from root node.
- Follow the path based on the attribute values of the given instance.
- Reach the leaf node → assign the class label.

Example : (Outlook = Sunny, Humidity = High, Windy = False)

Path : Outlook → Sunny → Humidity → High → Class = No

9. Advantages

- ✓ Easy to understand and interpret.
- ✓ Requires little data preparation.
- ✓ Handles both numerical and categorical data.
- ✓ Non-parametric (does not assume any distribution).

10. Limitations

- ✗ Can be unstable (small change in data may change the tree).
- ✗ May create overfitting if tree is too deep.
- ✗ Greedy approach → may not always give optimal tree.

In Short : Decision Tree Classification builds a tree by recursively splitting the data on the best attribute. It is simple, powerful and widely used for classification and prediction tasks.

Applications:

- Medical diagnosis
- Credit risk analysis
- Customer churn prediction
- Spam detection
- Fault diagnosis, etc.



Bayesian Classification

Bayesian Classification is a probabilistic technique based on Bayes' Theorem.

It uses prior knowledge (prior probability) and the observed data (likelihood) to compute the posterior probability of each class and assigns the instance to the class with the highest posterior probability.

1. Bayes' Theorem

Bayes' Theorem is given by:

$$P(C|X) = \frac{P(X|C) P(C)}{P(X)}$$

Where,

C : Class (hypothesis)
X : Data (attribute values / features)
P(C) : Prior probability of class C
P(X|C) : Likelihood (probability of X given class C)
P(C|X) : Posterior probability of class C given X
P(X) : Evidence (probability of X for all classes)

2. Naïve Bayes Assumption

Naïve Bayes classifier assumes that all attributes are conditionally independent given the class.

Thus,

$$P(X_1, X_2, \dots, X_n | C) = \prod_{i=1}^n P(X_i | C)$$

Where $X_1, X_2, \dots, X_n$ are attributes.

3. Classification Rule

Assign the instance X to the class C_k for which $P(C_k | X)$ is maximum.

$$P(C_k | X) = \frac{P(X | C_k) P(C_k)}{P(X)}$$

Since $P(X)$ is same for all classes, we maximize $P(X | C_k) P(C_k)$

4. Types of Bayesian Classifiers

- Gaussian Naïve Bayes**
 - Assumes continuous attributes follow normal (Gaussian) distribution.
- Multinomial Naïve Bayes**
 - Used for discrete data such as word counts (text classification).
- Bernoulli Naïve Bayes**
 - Used for binary / Boolean attributes (presence or absence).

5. Example - Play Tennis Dataset

ID	Outlook	Temp.	Humidity	Windy	PlayTennis
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rain	Mild	High	False	Yes
5	Rain	Cool	Normal	False	Yes
6	Rain	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rain	Mild	Normal	False	Yes

Goal:

For a new day with Outlook = Sunny, Temp = Cool, Humidity = High, Windy = True predict PlayTennis?

Step 1: Prior Probabilities

Class	Count	Prior Probability
Yes	6	$P(\text{Yes}) = 6/10 = 0.6$
No	4	$P(\text{No}) = 4/10 = 0.4$

6. Likelihood Calculation

For class = Yes (6 records)

Attribute	Value	Count	Likelihood $P(X_i \text{Yes})$
Outlook	Sunny	1	$1/6 = 0.167$
Temp.	Cool	3	$3/6 = 0.500$
Humidity	High	3	$3/6 = 0.500$
Windy	True	1	$1/6 = 0.167$

$$P(X | \text{Yes}) = 0.167 \times 0.500 \times 0.500 \times 0.167 = 0.00694$$

For class = No (4 records)

Attribute	Value	Count	Likelihood $P(X_i \text{No})$
Outlook	Sunny	2	$2/4 = 0.500$
Temp.	Cool	1	$1/4 = 0.250$
Humidity	High	4	$4/4 = 1.000$
Windy	True	2	$2/4 = 0.500$

$$P(X | \text{No}) = 0.500 \times 0.250 \times 1.000 \times 0.500 = 0.0625$$

7. Posterior Probability

For class = Yes:

$$P(\text{Yes} | X) \propto P(X | \text{Yes}) P(\text{Yes}) = 0.00694 \times 0.6 = 0.004164$$

For class = No:

$$P(\text{No} | X) \propto P(X | \text{No}) P(\text{No}) = 0.0625 \times 0.4 = 0.0250$$

Normalize:

$$P(X) = 0.004164 + 0.0250 = 0.029164$$

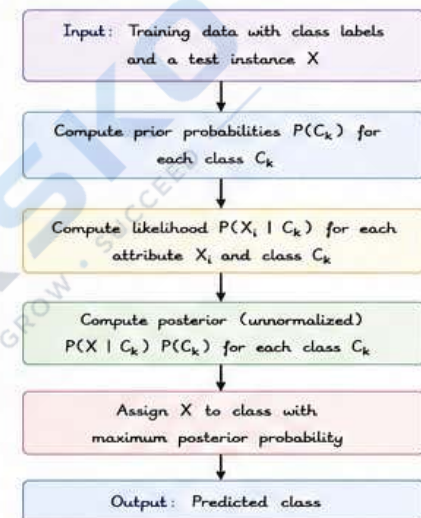
$$P(\text{Yes} | X) = \frac{0.004164}{0.029164} = 0.143$$

$$P(\text{No} | X) = \frac{0.0250}{0.029164} = 0.857$$

Prediction:

Since $P(\text{No} | X) > P(\text{Yes} | X)$, the day is predicted as **PlayTennis = No**

8. Algorithm (Naïve Bayes Classification)



9. Advantages

- Simple and easy to implement.
- Works well with high dimensional data.
- Requires small training data.
- Handles both categorical and continuous data (with Gaussian model).

Limitations

- Assumes independence between attributes which may not always hold.
- Zero probabilities may occur (can be solved using Laplace smoothing).
- Not as accurate as advanced models in complex datasets.

10. Dealing with Zero Probability - Laplace Smoothing

If $P(X_i | C_k) = 0$ (attribute value not seen in training data), use Laplace correction:

$$P(X_i = x | C_k) = \frac{\text{Count}(X_i = x, C_k) + 1}{\text{Count}(C_k) + m}$$

Where m = number of possible values of attribute X_i .

In Short

Bayesian Classification uses probability and Bayes' Theorem to classify data. It is simple, efficient and effective, especially for large datasets and text classification.



Rule-Based Classification

Rule-Based Classification is a technique that uses a set of **IF-THEN** rules derived from data or expert knowledge to classify new (**unseen**) instances into predefined classes.

1. What is Rule-Based Classification?

- It represents knowledge in the form of rules.
- Each rule has a condition (IF part) and a class label (THEN part).
- A new instance is classified by matching it against the rules.
- Easy to understand and implement.

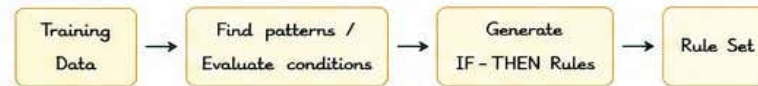
2. Structure of a Rule

IF (condition₁) AND (condition₂) AND ... AND (condition_n)
THEN Class = C_k [with optional confidence]

3. Example Dataset

ID	Outlook	Temp.	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes

4. How Rules are Generated?



Rules can be generated using:

- Decision Tree
- Association Rules
- Inductive Logic Programming
- Expert Knowledge

5. Example Rule Set (from above dataset)

R1: IF Outlook = Overcast THEN PlayTennis = Yes
R2: IF Outlook = Sunny AND Humidity = High THEN PlayTennis = No
R3: IF Outlook = Sunny AND Humidity = Normal THEN PlayTennis = Yes
R4: IF Outlook = Rain AND Wind = Strong THEN PlayTennis = No
R5: IF Outlook = Rain AND Wind = Weak THEN PlayTennis = Yes

6. Classifying a New Instance

New Instance	
Outlook	Sunny
Temp.	Cool
Humidity	High
Wind	Weak
Play Tennis	?

Match with Rules (in order)

R1: Outlook = Overcast ? → No
R2: Outlook = Sunny AND Humidity = High ? → Yes ✓

=> Class = No

7. Resolving Conflicts

If more than one rule matches, use:

- Priority Order (rule order)
- Highest Confidence
- Most Specific Rule (more conditions)
- Majority Vote / Weighted Vote

8. Types of Rule-Based Classifiers

Type	Description
Separate-and-Conquer	Sequentially learns one rule at a time covering some training examples.
Covering Algorithm	Learns rules until all (or most) examples are covered.
Ordered Rule List	Rules are applied in a specific order; first matching rule is used.
Unordered Rule Set	All matching rules are considered; class decided by voting or accuracy.

9. Advantages

- ✓ Easy to understand and interpret.
- ✓ Works well with both numerical and categorical data.
- ✓ Handles missing values (can use rules with available values).
- ✓ Fast classification once rules are built.
- ✓ Can incorporate expert knowledge.

10. Limitations

- ✗ May produce a large number of rules.
- ✗ Rule order may affect the result.
- ✗ Can be less accurate if rules are inconsistent or incomplete.
- ✗ Not suitable for highly complex patterns.

11. General Process



12. Applications

- Medical diagnosis
- Credit scoring
- Customer segmentation
- Spam detection
- Fault diagnosis
- Fraud detection



In Short : Rule-Based Classification uses a set of IF-THEN rules to assign class labels.

It is simple, interpretable and effective, especially when the knowledge can be expressed as rules.

Accuracy and Error Measures

Accuracy and Error Measures are used to evaluate the performance of a classification or prediction model. They compare the predicted results with the actual (true) values.

1. Confusion Matrix

It is a table that describes the performance of a classification model on a set of test data for which the true values are known.

Actual vs Predicted

	Predicted Class	
	Positive (P)	Negative (N)
Actual (True) Class		
Positive (P)	True Positive (TP)	False Negative (FN)
Negative (N)	False Positive (FP)	True Negative (TN)

Where,

TP : Predicted Positive and Actual Positive (Correct Positive)

TN : Predicted Negative and Actual Negative (Correct Negative)

FP : Predicted Positive but Actual Negative (Type I Error)

FN : Predicted Negative but Actual Positive (Type II Error)

2. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

It is the proportion of total correct predictions.

Example :

If TP = 50, TN = 30, FP = 10, FN = 10

$$\text{Accuracy} = \frac{50 + 30}{50 + 30 + 10 + 10} = \frac{80}{100} = 0.80 \text{ (80\%)}$$

3. Error (Misclassification) Rate

$$\text{Error Rate} = \frac{FP + FN}{TP + TN + FP + FN} = 1 - \text{Accuracy}$$

Example :

$$\text{Error Rate} = \frac{10 + 10}{50 + 30 + 10 + 10} = \frac{20}{100} = 0.20 \text{ (20\%)}$$

4. Other Common Performance Measures

• Precision (Positive Predictive Value)

$$\text{Precision} = \frac{TP}{TP + FP}$$

Out of all instances predicted Positive, how many are actually Positive.

• Recall (Sensitivity, True Positive Rate)

$$\text{Recall} = \frac{TP}{TP + FN}$$

Out of all actual Positive instances, how many are correctly predicted.

• Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Out of all actual Negative instances, how many are correctly predicted.

• F1 - Score

$$\text{F1 - Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean of Precision and Recall. Useful when classes are imbalanced.

5. Measure in Terms of Error

• False Positive Rate (FPR) / Fall-out

$$\text{FPR} = \frac{FP}{FP + TN}$$

Proportion of actual Negative instances incorrectly predicted as Positive.

• False Negative Rate (FNR) / Miss Rate

$$\text{FNR} = \frac{FN}{FN + TP}$$

Proportion of actual Positive instances incorrectly predicted as Negative.

• False Discovery Rate (FDR)

$$\text{FDR} = \frac{FP}{TP + FP}$$

Proportion of positive predictions that are actually negative.

6. Summary Table of Measures

Measure	Formula	Best Value	Meaning
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	1 (or 100%)	Overall correctness
Error Rate	$\frac{FP + FN}{TP + TN + FP + FN}$	0 (or 0%)	Overall incorrectness
Precision	$\frac{TP}{TP + FP}$	1	Correctness of Positive predictions
Recall (Sensitivity)	$\frac{TP}{TP + FN}$	1	Ability to find all Positive instances
Specificity	$\frac{TN}{TN + FP}$	1	Ability to find all Negative instances
F1 - Score	$\frac{2PR}{P + R}$	1	Balance of Precision and Recall

7. Example - Complete Calculation

Confusion Matrix

	Predicted	
	Positive	Negative
Actual		
Positive	50	10
Negative	10	30

Higher values of Accuracy, Precision, Recall, Specificity and F1-Score indicate better model performance, while lower values of Error Rate, FPR and FNR indicate better performance.

- Accuracy = $(50 + 30) / (50 + 30 + 10 + 10) = 80 / 100 = 0.80 \text{ (80\%)}$
- Error Rate = $(10 + 10) / 100 = 0.20 \text{ (20\%)}$
- Precision = $50 / (50 + 10) = 0.833 \text{ (83.3\%)}$
- Recall = $50 / (50 + 10) = 0.833 \text{ (83.3\%)}$
- Specificity = $30 / (30 + 10) = 0.75 \text{ (75\%)}$
- FPR = $10 / (10 + 30) = 0.25 \text{ (25\%)}$
- FNR = $10 / (10 + 50) = 0.167 \text{ (16.7\%)}$
- F1 - Score = $2 \times 0.833 \times 0.833 / (0.833 + 0.833) = 0.833 \text{ (83.3\%)}$

In Short :

➤ Accuracy tells how many predictions are correct overall.

➤ Precision tells how many predicted positives are actually positive.

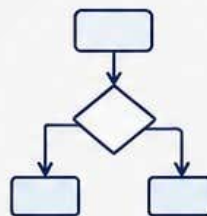
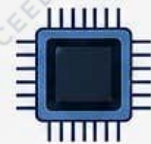
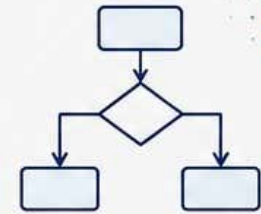
➤ Recall tells how many actual positives are found.

➤ Error measures (FPR, FNR, Error Rate) tell how many mistakes are made.

✗ Use appropriate measures depending on the problem, especially when the dataset is imbalanced.



1010101010
0101010101
1010101010
0101010101



Unit - III



10110
01001
10101



```
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello, World!";
    return 0;
}
```

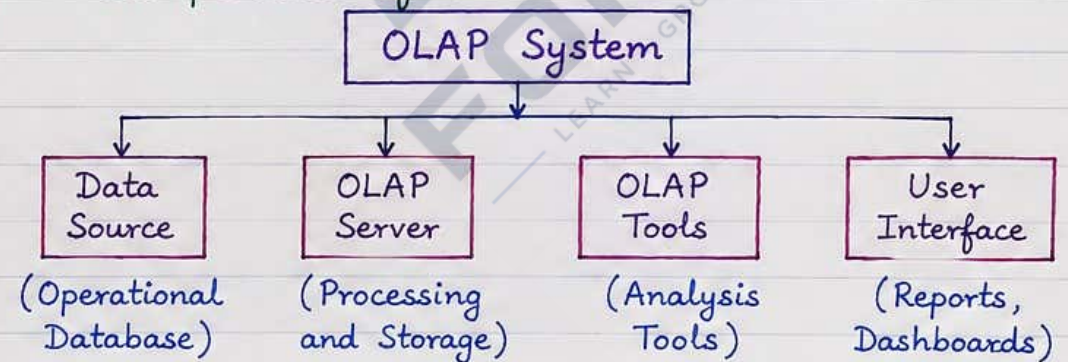

Clustering and OLAP Technology : Overview Of Technology

1) Clustering

- Clustering is an unsupervised data mining technique that groups similar objects together in the same cluster and dissimilar objects in different clusters.
- Purpose :
To discover natural groupings or patterns in data without using predefined classes.
- Characteristics :
 - No prior knowledge of class labels.
 - High intra-cluster similarity.
 - Low inter-cluster similarity.
 - Useful for data exploration, segmentation, and summarization.
- Examples :
Customer segmentation, Market basket analysis, Document clustering, Image segmentation, etc.
- In short : Clustering is used to group similar data, while OLAP is used to analyze multidimensional data for better decision making.

2) OLAP Technology

- OLAP (Online Analytical Processing) is a technology used for multi-dimensional data analysis to support decision making.
- Purpose :
To provide fast, consistent, interactive analysis of large volumes of data from multiple perspectives.
- Key Features :
 - Multidimensional data view.
 - Fast query response.
 - Supports aggregation and summarization.
 - Used in decision support systems (DSS).
- Components of OLAP :



Data Cube and Data Generalization

1) Data Cube :

- A data cube is a multidimensional structure that organizes data along multiple dimensions and measures to support OLAP (Online Analytical Processing).
- It allows data to be viewed and analyzed from different perspectives by performing operations such as roll-up, drill-down, slice and dice.
- Each cell in the cube represents a value of a measure for a combination of dimension values.

Example : Sales Data Cube

Dimensions → Time, Product, Location

Measure → Sales (in ₹)

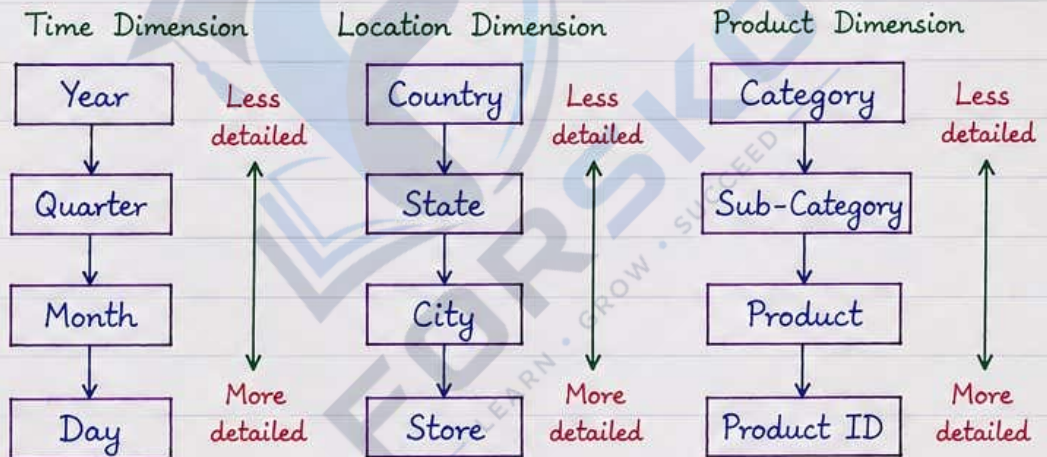
Time	Product			Location
	TV	Mobile	Laptop	
	2024 Q2	120	150	100
	2024 Q1	100	130	90
	2023 Q2	90	110	80
	2021	80	100	70

North
West
South

2) Data Generalization :

- Data generalization is the process of replacing detailed data with higher-level concepts in a dimension hierarchy.
- It reduces the granularity of data and helps in data summarization.
- Used in operations like roll-up in OLAP.

Example : Generalization Hierarchy



Use :

- Generalization helps in summarizing data at higher levels.
- It reduces the size of the data to be processed.
- Improves query performance and supports decision making.

Introduction to Clustering

1) What is Clustering ?

- Clustering is an **unsupervised learning** technique used to group similar objects together into clusters.
- Objects within a cluster are **more similar** to each other and **less similar** to objects in other clusters.
- It discovers **natural groupings** or **patterns** in data without any predefined class labels.

2) Purpose / Objectives

- To discover inherent group structure in data.
- To summarize large datasets.
- To segment data for better understanding.
- To detect outliers.
- To help in decision making and data analysis.

3) Characteristics

- Works on unlabeled data (unsupervised).
- Based on similarity / distance between objects.
- Different clustering algorithms may produce different results.
- The number of clusters may be known in advance or determined automatically.

4) Types of Data Suitable for Clustering

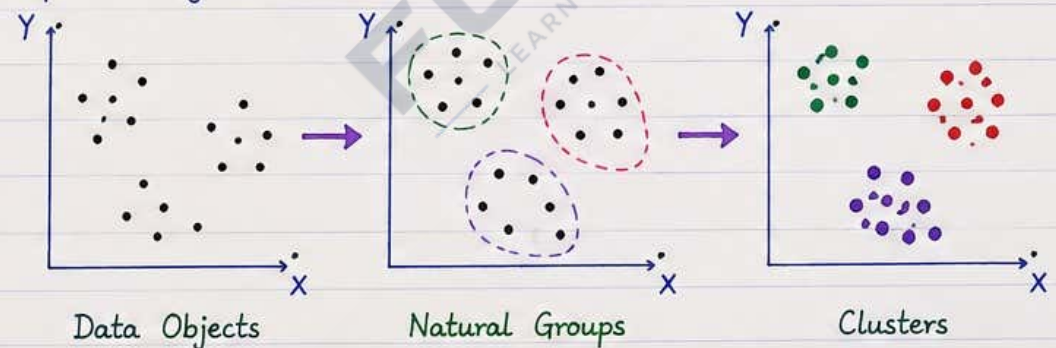
- **Numeric data** (e.g., age, income, temperature)
- **Categorical data** (e.g., color, region)
- **Mixed data** (combination of numeric and categorical)
- **High-dimensional data**

5) Applications

- Market segmentation
- Customer grouping
- Document clustering
- Image segmentation
- Social network analysis
- Bioinformatics, Web usage mining, etc.

6) Example

Given a set of data points, clustering groups similar points together.



In short : Clustering groups similar data together without using any prior knowledge of class labels.

Partitioning Methods

1) Introduction :

Partitioning methods divide a dataset of n objects into k ($k < n$) partitions or clusters such that the objects within a cluster are more similar (homogeneous) to each other and less similar (heterogeneous) to objects in other clusters.

2) Basic Idea :

- Given : A set of n objects and number of clusters k .
- Find : A partition of the objects into k clusters.
- Objective : Optimize a criterion function (usually minimize intra-cluster distance).

3) Common Algorithm : k-Means Algorithm

It is the most popular partitioning method.

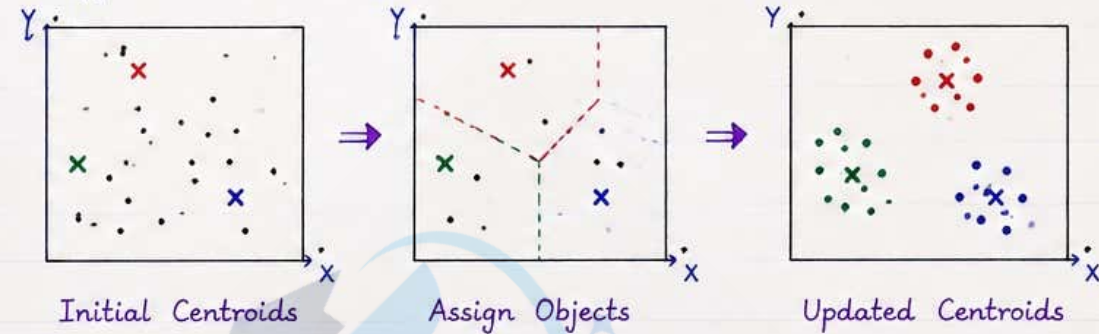
Steps :

- ① Choose the number of clusters k .
- ② Select k initial centroids (randomly or using some heuristic).
- ③ Assign each object to the nearest centroid (based on distance, e.g., Euclidean distance).
- ④ Recompute the centroid of each cluster.
- ⑤ Repeat steps ③ and ④ until no change in cluster assignments or centroids (convergence).

In Short : Partitioning methods (e.g., k-means) partition data into k clusters by optimizing a criterion such as minimizing within-cluster distance.

4) Example :

Suppose $k = 3$ clusters.



5) Objective Function :

Minimize the total within-cluster sum of squares (WCSS) :

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} d(x, \mu_i)^2$$

where, C_i = i^{th} cluster

μ_i = centroid (mean) of cluster C_i

$d(x, \mu_i)$ = distance between object x and centroid μ_i

6) Characteristics :

- Simple and efficient.
- Works well for spherical, compact clusters.
- Sensitive to initial centroid selection.
- Requires the number of clusters (k) as input.

Hierarchical Methods

1) Introduction :

Hierarchical methods create a hierarchy (tree) of clusters. It produces a nested sequence of clusters organized as a hierarchy. The result is represented by a tree diagram called dendrogram.

2) Types of Hierarchical Methods :

- **Agglomerative (Bottom-Up)**: Start with each object as a separate cluster and successively merge the closest clusters until all objects are in a single cluster.
- **Divisive (Top-Down)**: Start with all objects in one cluster and successively split clusters until each object forms a separate cluster.

3) Agglomerative Hierarchical Clustering :

- Each object is initially a cluster.
- Compute the distance (similarity) between clusters.
- Merge the two closest clusters.
- Update the distance matrix.
- Repeat steps 2-4 until only one cluster remains.

Linkage Criteria (ways to compute inter-cluster distance):

- **Single Link (Min Link)** : Minimum distance between any two objects in the clusters.
- **Complete Link (Max Link)** : Maximum distance between any two objects in the clusters.
- **Average Link (UPGMA)** : Average of all pairwise distances between objects in clusters.
- **Centroid Link** : Distance between centroids (means) of the clusters.

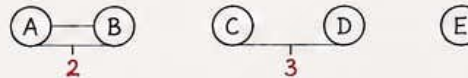
4) Agglomerative Example :

Consider 5 objects : A, B, C, D, E
(numbers indicate distance at which clusters are merged)

Step 0
(Initial)



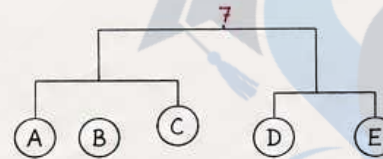
Step 1



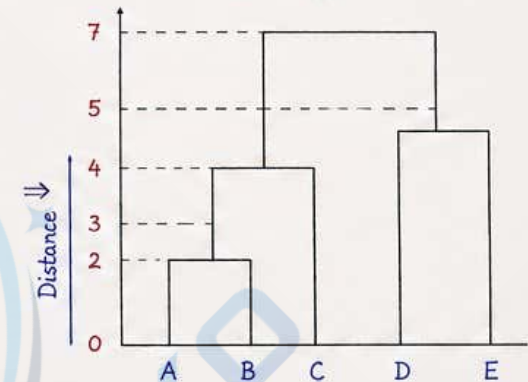
Step 2



Step 3

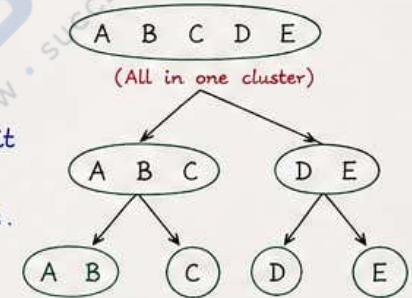


Dendrogram



5) Divisive Hierarchical Clustering :

- Start with all objects in one cluster.
- Find the most dissimilar pair (or cluster) and split the cluster.
- Repeat the process recursively on resulting clusters.
- Stop when each object forms its own cluster.



6) Characteristics :

- Does not require the number of clusters as input.
- Produces a dendrogram (visual representation of cluster hierarchy).
- More computationally expensive : $O(n^2)$ time and space.
- Sensitive to noise and outliers.

In Short : Hierarchical methods build a hierarchy of clusters either by merging (agglomerative) or splitting (divisive) and are represented using a dendrogram.

Density-Based Methods

1) Introduction :

Density-based clustering methods group together objects that are closely packed together (high density) and mark objects that lie in low density regions as outliers (noise).

2) Basic Idea :

- A cluster is defined as a dense region of points that is separated by regions of low density.
- These methods can discover clusters of arbitrary shape.
- They can identify noise (outliers) automatically.

3) Key Concepts :

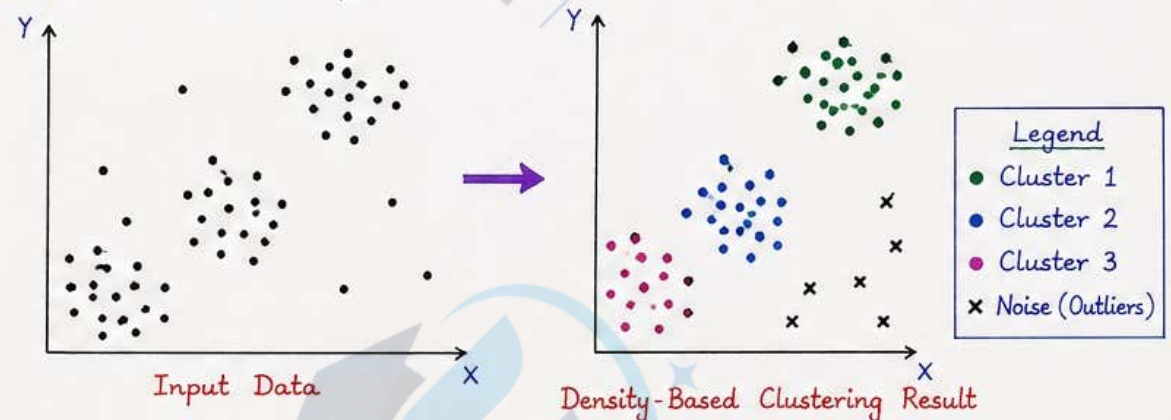
- ϵ (Epsilon) : Radius of the neighborhood.
- MinPts : Minimum number of points required within ϵ -neighborhood to form a dense region.
- Core Point : A point having at least MinPts points within its ϵ -neighborhood (including itself).
- Border Point : A point that is within ϵ -neighborhood of a core point but has less than MinPts points in its own ϵ -neighborhood.
- Noise Point : A point that is neither a core nor a border point.

4) Algorithm (DBSCAN) :

- ① Choose parameters ϵ (radius) and MinPts.
- ② Pick an unvisited point p .
- ③ Retrieve all points within ϵ -neighborhood of p .
→ If number of points $<$ MinPts, label p as Noise.
→ Otherwise, create a new cluster and mark p as core point.
- ④ Recursively visit all points density-reachable from p (i.e., within ϵ of core points) and add them to the cluster.
- ⑤ Repeat steps ② to ④ for all unvisited points.

5) Example :

Given $\epsilon = 1.5$ units, MinPts = 4



6) Properties / Characteristics :

- Finds clusters of arbitrary shape.
- Handles noise (outliers) effectively.
- Requires two input parameters: ϵ and MinPts.
- Suitable for large datasets.
- Results are sensitive to the choice of ϵ and MinPts.

7) Common Density-Based Algorithms :

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- OPTICS (Ordering Points To Identify the Clustering Structure)
- DENCLUE (DENsity-based CLUstEring)

Difference from Other Methods :

- Unlike partitioning methods (e.g., k-means), no need to specify the number of clusters.
- Unlike hierarchical methods, does not require building a dendrogram.

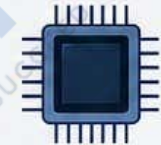
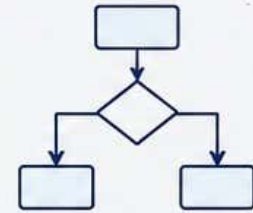
In Short : Density-based methods find dense regions of objects separated by low-density regions. They discover clusters of arbitrary shape and can detect noise automatically.

Difference between Classification and Clustering

Aspect	Classification	Clustering
1. Definition	Classification is a supervised learning technique that assigns objects to predefined classes (or categories).	Clustering is an unsupervised learning technique that groups similar objects together into clusters.
2. Learning Type	Supervised Learning	Unsupervised Learning
3. Input Data	Uses labeled (classified) training data with known class labels.	Uses unlabeled data with no prior information about classes.
4. Goal	To predict the class label of new / unseen data based on learned model.	To discover natural groupings or patterns in data.
5. Output	Assigns each object to one of the predefined classes.	Forms clusters such that objects in the same cluster are similar and objects in different clusters are dissimilar.
6. Number of Groups	Determined in advance (fixed number of classes).	Not known in advance; determined by the algorithm.
7. Purpose	Used for prediction and decision making.	Used for exploration and pattern discovery.
8. Algorithms (Examples)	Decision Trees, Naïve Bayes, SVM, K-Nearest Neighbor, Logistic Regression etc.	K-Means, Hierarchical Clustering, DBSCAN, Fuzzy C-Means, etc.
9. Evaluation	Performance is measured using accuracy, precision, recall, F1-score, etc.	Evaluation is difficult; uses measures like Silhouette Coefficient, Davies-Bouldin Index, Within-Cluster Sum of Squares, etc.
10. Typical Use Cases	Email spam detection, Disease diagnosis, Credit risk prediction, Image classification, etc.	Customer segmentation, Market basket analysis, Document grouping, Gene expression analysis, etc.
11. Example	Predict whether an email is "Spam" or "Not Spam".	Group customers into clusters based on their purchasing behavior.

In Short : Classification uses known data to predict class labels, while clustering uses unknown data to find natural groups or clusters.

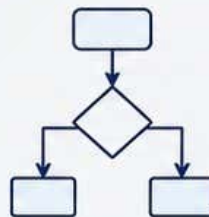
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Unit - IV



10110
01001
10101



```
#include <iostream>
using namespace std;
int main()
{
    cout << "Hello, World!";
    return 0;
}
```


Advanced Topics and Applications :

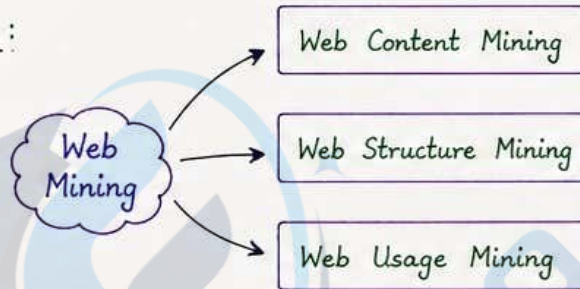
Introduction to Web Mining and Text Mining

1) Web Mining :

Web Mining is the application of data mining techniques to discover useful patterns and knowledge from Web data and Web usage.

a) Sources of Web Data :

- Web Content
- Web Structure
- Web Usage (Log) Data



b) Types of Web Mining :

- **Web Content Mining :** Extracts useful information from the content of Web pages.
- **Web Structure Mining :** Analyzes the link structure of Web pages (e.g., PageRank, HITS algorithm).
- **Web Usage Mining :** Discovers patterns from Web logs (user access behavior).

c) Applications of Web Mining :

- Personalization of Web services
- E-commerce and recommendation
- Web advertisement and marketing
- Website improvement and management
- Fraud detection

Example :

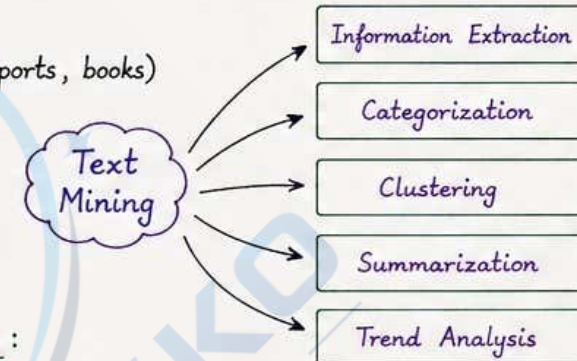
Analyzing user navigation patterns on an e-commerce site to recommend products that the user is likely to buy.

2) Text Mining :

Text Mining is the process of discovering interesting and non-trivial patterns or knowledge from unstructured text documents.

a) Sources of Text Data :

- Documents (e.g., articles, reports, books)
- Emails
- Social media posts
- News and blogs
- Reviews and feedback



b) Key Tasks in Text Mining :

- **Information Retrieval :** Finding relevant documents.
- **Text Classification :** Assigning documents to predefined categories.
- **Clustering :** Grouping similar documents.
- **Information Extraction :** Extracting entities, events, relationships.
- **Summarization :** Creating concise summary of documents.
- **Sentiment Analysis :** Determining opinion or sentiment in text.

c) Applications of Text Mining :

- Spam email filtering
- Customer feedback analysis
- News trend detection
- Document organization and management
- Market and opinion analysis

Example :

Analyzing customer reviews of a product to find common issues and overall sentiment.

In Short : Web Mining focuses on discovering patterns from Web data (content, structure, usage), while Text Mining focuses on extracting meaningful information from textual data.

Social Network Analysis (SNA)

1) Introduction :

Social Network Analysis (SNA) is the process of studying relationships and interactions among actors (people, groups, organizations, etc.) in a network. It helps to discover patterns, influential entities, communities and the structure of relationships.

2) Basic Concepts :

- **Actor (Node)** : An individual entity in the network.
- **Tie (Edge)** : Relationship or interaction between two actors.
- **Network** : A set of actors (nodes) and ties (edges).
- **Graph** : Mathematical representation of a network $G(V, E)$ where V = nodes, E = edges.

3) Types of Social Networks :

- **Directed Network** : Relationships have direction (e.g., follower → following).
- **Undirected Network** : Relationships are mutual (e.g., friendship).
- **Weighted Network** : Ties have strength/weight (e.g., frequency of communication).
- **Unweighted Network** : All ties are considered equal.

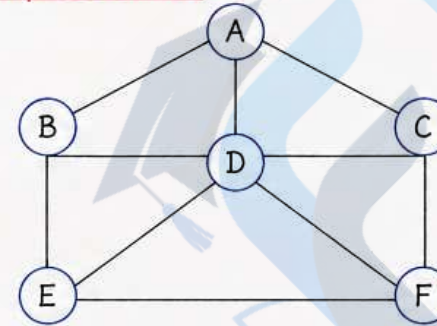
4) Elements of a Social Network :

Element	Description
Actors (Nodes)	Individuals or entities in the network.
Ties (Edges)	Relationships or interactions between actors.
Attributes	Properties of actors (age, role, location, etc.).
Communities	Groups of actors more densely connected to each other.
Subgraphs	Smaller parts of the network.

5) Key Measures / Metrics in SNA :

- **Degree Centrality** : Number of connections an actor has.
- **Betweenness Centrality** : How often an actor lies on the shortest path between other actors.
- **Closeness Centrality** : How close an actor is to all other actors.
- **Eigenvector Centrality** : Influence of an actor considering the importance of its connections.
- **Density** : Proportion of actual ties to possible ties in the network.

6) Example Network :



Example Measures :

Degree(A) = 3
Degree(D) = 4
Betweenness(D) = High
Closeness(A) = High
Density = $10 / 15 = 0.67$
(Undirected network)

7) Applications of SNA :

- Identifying key influencers in a network.
- Detecting communities and groups.
- Analyzing information diffusion.
- Recommender systems.
- Fraud detection.
- Organizational analysis.
- Understanding social behavior and trends.

8) Tools for Social Network Analysis :

Gephi, UCINET, NodeXL, Pajek, NetworkX (Python), R (igraph package) etc.

★ NetworkX Example (Python)

```
import networkx as nx
G = nx.Graph()
G.add_edges_from([('A', 'B'),
                  ('A', 'C'),
                  ('B', 'D'), ...])
nx.degree_centrality(G)
```

In Short : Social Network Analysis helps to model, visualize and analyze relationships in a network. It reveals structure, important actors and communities to support decision making.

GRAPH MINING

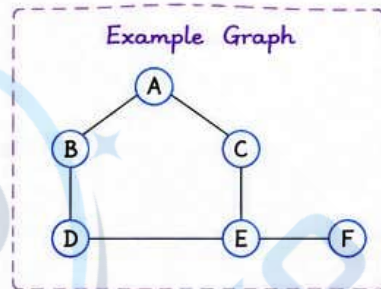
Graph Mining is the process of discovering interesting, useful and previously unknown patterns from graph-structured data.

1) Introduction :

- Graph mining applies data mining techniques to extract patterns from graphs.
- A graph $G = (V, E, L_v, L_e)$ consists of a set of vertices (V), edges (E) and optional vertex labels (L_v) and edge labels (L_e).

2) Graph Data Sources :

- Social networks (e.g., Facebook, Twitter)
- Web graphs (web pages and hyperlinks)
- Biological networks (protein-protein interaction)
- Chemical structures
- Communication and transportation networks
- Citation networks



3) Types of Graphs :

- Undirected Graph** : Edges have no direction.
- Directed Graph (Digraph)** : Edges have direction.
- Labeled Graph** : Vertices and/or edges have labels.
- Attributed Graph** : Vertices/edges have attributes (values).
- Weighted Graph** : Edges have weights or costs.
- Attributed & Graph** : Both structure and attributes are present.

4) Types of Patterns in Graph Mining :

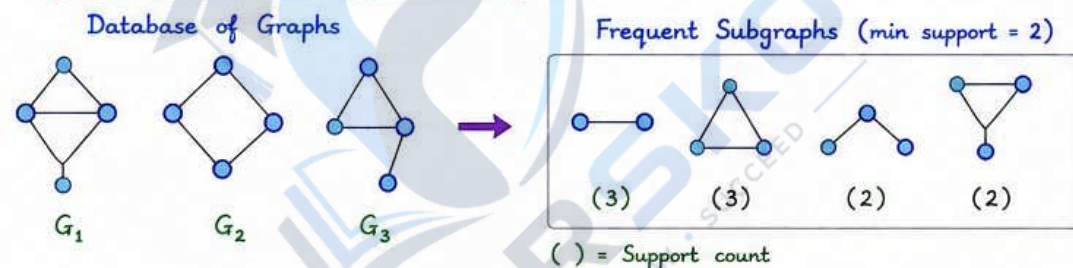
Frequent Subgraph (Structural Pattern)	Graph Motif	Subgraph Isomorphism	Graph Classification	Graph Clustering
Triangle is a frequent subgraph.	Small, recurring subgraph.	Check if a pattern exists in a graph.	Classify graphs into categories.	Group similar graphs together.

In Short : Graph Mining discovers meaningful patterns from graph-structured data such as communities, important substructures, and similarity between graphs, which helps in better decision making and knowledge discovery.

5) Graph Mining Tasks :

Task	Description
Subgraph Discovery	Find frequent or interesting subgraphs in a large graph.
Pattern Matching	Find all occurrences of a query pattern in a graph.
Graph Classification	Classify graphs into predefined classes.
Graph Clustering	Group similar graphs based on structure or attributes.
Outlier / Anomaly Detection	Detect rare or unusual graphs or substructures.
Link Prediction	Predict missing links or future connections.
Graph Summarization	Generate a compact summary of a large graph.

6) Example - Frequent Subgraph Mining



7) Applications of Graph Mining :

- Social Network Analysis** : community detection, influence analysis, friend recommendation.
- Bioinformatics** : protein interaction networks, gene regulatory networks.
- Web Mining** : link analysis, page ranking, web community detection.
- chemoinformatics** : finding functional groups, drug discovery.
- Fraud Detection** : identifying suspicious groups and transactions.

8) Challenges :

- Huge size of graphs (vertices/edges).
- Complex and dynamic structure.
- NP-hard nature of many graph problems.
- Lack of standard representations.

Common Graph Mining Algorithms

- GSpan - frequent subgraph mining
- FANMOD - frequent subgraph mining
- Graph Classification - Graph Kernels, GNNs
- Graph Clustering - Spectral clustering, MCL
- Link Prediction - Common Neighbors, Adamic/Adar

Mining Time-Series Data

Time-series data are observations recorded over time at regular or irregular intervals.
Mining time-series data aims to discover useful patterns, trends and relationships.

1) Introduction

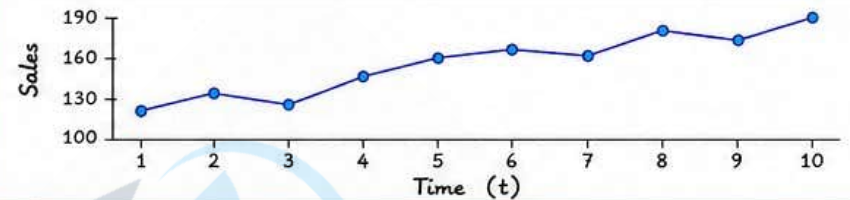
- Time-series data are sequences of data points indexed (or ordered) in time order.
- Examples: stock prices, sensor readings, weather data, ECG signals, web traffic, electricity demand, etc.
- Time adds a dynamic aspect - the order of observations is important.

2) Characteristics of Time-Series Data

- **Time dependency** (correlation between nearby observations)
- **Trends** (long-term increase/decrease)
- **Seasonality** (regular patterns at fixed periods)
- **Noise and outliers**
- **Non-stationarity** (statistical properties change over time)

3) Example of Time-Series Data

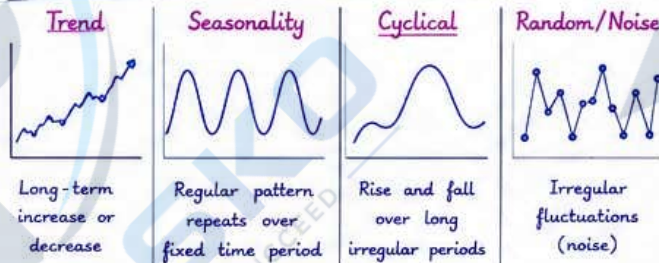
Time (t)	1	2	3	4	5	6	7	8	9	10
Sales	120	132	128	143	155	160	158	170	165	180



4) Types of Time-Series Data

- **Univariate Time Series**: Single variable observed over time.
- **Multivariate Time Series**: Multiple variables observed over time.
- **Irregularly Sampled Series**: Observations at non-constant time intervals.
- **Time Series with Missing Values**: Some observations are missing.

5) Common Patterns in Time-Series



6) Mining Tasks on Time-Series Data

- **Similarity Search**: Find time series most similar to a query series.
- **Clustering**: Group similar time series.
- **Classification**: Assign a class label to a time series.
- **Trend/Pattern Discovery**: Discover trends, seasonality, recurring patterns.
- **Anomaly/Outlier Detection**: Detect unusual or rare events.
- **Forecasting/Prediction**: Predict future values.
- **Motif Discovery**: Find frequently occurring subsequences.

7) Common Techniques

- **Smoothing**: Remove noise (e.g., moving average, exponential smoothing).
- **Normalization**: Remove scale/offset effects.
- **Discretization**: Convert continuous values into discrete symbols.
- **Dimensionality Reduction**: PCA, DFT, DWT, PAA (Piecewise Aggregate Approximation).
- **Distance/Similarity Measures**: Euclidean, DTW (Dynamic Time Warping), LCSS, Correlation-based, etc.

8) Example - Similarity using DTW

Query Series (Q)



Candidate Series (C)



DTW aligns two series by warping the time axis to find the optimal match.

DTW Distance (Q, C) = 2.35

9) Applications

- **Finance**: Stock price analysis, risk monitoring, portfolio management.
- **Business**: Sales forecasting, demand prediction.
- **Healthcare**: ECG analysis, patient monitoring.
- **IoT & Sensors**: Fault detection, predictive maintenance.
- **Weather & Climate**: Temperature, rainfall pattern analysis.
- **Web Analytics**: User behavior over time.

10) Challenges

- High volume and velocity of data.
- Noise, missing values and outliers.
- Scalability and computational complexity.
- Non-stationarity and evolving patterns.
- Choosing appropriate similarity measures.

Example: Missing Values

Original Series :	4	6	-	10	-	14	16
	t ₁	t ₂	t ₃	t ₄	t ₅	t ₆	t ₇
After Interpolation :	4	6	8	10	12	14	16
	t ₁	t ₂	t ₃	t ₄	t ₅	t ₆	t ₇

11) Evaluation Measures

- For **Similarity Search**: Precision, Recall
- For **Classification**: Accuracy, F1-score
- For **Clustering**: Silhouette Coefficient, Purity
- For **Forecasting**: MAE, RMSE, MAPE

In Short

Mining time-series data helps to discover meaningful patterns, make accurate predictions and support better decision making in dynamic environments where time plays a crucial role.

Applications Of Data Mining in Healthcare, Banking

Data mining extracts useful **patterns** and **knowledge** from large datasets. It helps organizations to make **better decisions**, improve **efficiency** and provide **personalized services**.



1. Applications Of Data Mining in Healthcare

Data mining techniques help in analyzing medical data to improve patient care, reduce costs and support decision making.

Application	Description	Benefits
①  Disease Prediction	Predict the likelihood of diseases (e.g., diabetes, heart disease, cancer) using patient history, symptoms, lab reports.	• Early detection • Timely treatment • Improved survival rate
②  Patient Segmentation	Group patients with similar characteristics or conditions.	• Personalized treatment • Targeted marketing • Better resource allocation
③  Treatment Effectiveness Analysis	Analyze treatment outcomes and identify which treatments are most effective for specific patient groups.	• Better clinical decisions • Higher success rate • Reduced side effects
④  Hospital Management	Analyze data on bed occupancy, patient flow, staff performance and resource utilization.	• Cost reduction • Efficient operations • Improved patient satisfaction
⑤  Drug Discovery and Development	Identify patterns in biological data to discover new drugs and understand drug responses.	• Faster drug discovery • Reduced research cost • Better drug safety
⑥  Fraud and Abuse Detection (in Health Insurance)	Detect unusual claims or billing patterns to identify fraud and abuse.	• Prevent financial loss • Improved claim accuracy • Better compliance
⑦  Medical Image Analysis	Use data mining with image patterns (X-ray, MRI, CT scan) for diagnosis support.	• Accurate diagnosis • Early detection • Support for radiologists

Example: Predicting the risk of heart disease using patient age, blood pressure, cholesterol level, ECG results and family history.



2. Applications Of Data Mining in Banking

Data mining helps banks to understand customers better, manage risks, prevent fraud and increase profitability.

Application	Description	Benefits
①  Customer Segmentation	Group customers based on behavior, income, transaction patterns, demographics, etc.	• Targeted marketing • Personalized offers • Higher customer satisfaction
②  Credit Scoring and Risk Assessment	Predict the creditworthiness of loan applicants using past loan data, income, assets, etc.	• Better lending decisions • Reduced bad loans • Lower default rate
③  Fraud Detection	Identify unusual or suspicious transactions using patterns and anomalies.	• Prevent financial loss • Real-time alert • Improved security
④  Market Basket Analysis	Find products/services commonly used together (e.g., loans with insurance).	• Cross-selling • Increased revenue • Better product bundling
⑤  Customer Churn Prediction	Predict customers who are likely to leave the bank.	• Customer retention • Reduced churn rate • Increased loyalty
⑥  Sales and Profitability Analysis	Analyze customer profitability and product performance.	• Focus on profitable customers • Optimize pricing • Improved profitability
⑦  Compliance and Regulatory Reporting	Analyze transactions to detect money laundering and ensure regulatory compliance.	• Regulatory compliance • Avoid penalties • Enhanced reputation

Example: Detecting fraudulent credit card transactions by analyzing location, amount, time and spending behavior of customers.



In Short:

In Healthcare - Data mining improves patient care, optimizes resources and reduces costs.

In Banking - Data mining increases customer satisfaction, reduces risk and improves profitability.

Applications of Data Mining in Education and Business

Data mining discovers useful patterns and knowledge from educational and business data to support decision making, improve performance, optimize processes and provide better services.



1. Applications of Data Mining in Education

Data mining helps educational institutions understand students better, improve learning outcomes and make data-driven decisions.

Application	Description	Benefits
①  Student Performance Prediction	Predict students' academic performance using data such as attendance, assignments, grades, quizzes, etc.	• Early identification of at-risk students • Timely intervention • Improved academic success
②  Student Segmentation (Clustering)	Group students with similar characteristics, behavior or performance.	• Personalized learning • Targeted support • Better resource allocation
③  Learning Analytics	Analyze learning behavior and engagement data from LMS, online courses, assessments, etc.	• Understand learning patterns • Improve course design • Enhance student engagement
④  Dropout Prediction	Identify students who are likely to drop out based on historical and behavioral data.	• Reduce dropout rate • Retention strategies • Better student support
⑤  Course Recommendation	Recommend courses or learning resources based on students' interests and past performance.	• Personalized learning path • Increased enrollment in relevant courses • Higher satisfaction
⑥  Resource Utilization Optimization	Analyze data on classrooms, facilities and faculty usage.	• Efficient scheduling • Better infrastructure planning • Cost optimization

Example: Predicting students at risk of failing in a semester using attendance, assignment submission and internal test scores to provide timely support.



2. Applications of Data Mining in Business

Data mining helps businesses understand customers, improve operations, reduce costs and increase profitability.

Application	Description	Benefits
①  Customer Segmentation	Group customers based on behavior, demographics, purchase history, income, etc.	• Targeted marketing • Personalized offers • Higher customer satisfaction
②  Market Basket Analysis	Find items frequently purchased together.	• Product bundling • Cross-selling • Increased sales
③  Sales Forecasting	Predict future sales using historical data, trends and seasonal patterns.	• Better demand planning • Optimal inventory • Increased profitability
④  Customer Churn Prediction	Identify customers likely to stop using the product/service.	• Improve customer retention • Reduce churn rate • Increase customer lifetime value
⑤  Fraud Detection	Detect unusual or fraudulent transactions or activities.	• Prevent financial loss • Real-time alerts • Improved security
⑥  Operational Efficiency	Analyze business processes and identify bottlenecks or inefficiencies.	• Cost reduction • Process improvement • Higher productivity
⑦  Product / Service Recommendation	Recommend products or services based on customer preferences and behavior.	• Increased sales • Better user experience • Customer loyalty

Example: Using past purchase data to recommend relevant products to customers on an e-commerce website, increasing sales and customer satisfaction.



In Education: Data mining improves student learning, reduces dropouts and helps institutions make smarter, data-driven decisions.



In Business: Data mining improves customer understanding, optimizes processes, reduces costs and increases profitability.

Recent Trends in Data Mining

Data mining is continuously evolving to handle new types of data, solve complex problems and provide real-time, accurate and intelligent solutions.

- ① **Big Data Mining** : Handling very large volume, velocity and variety of data using distributed frameworks. → **Example:** Web logs analysis, clickstream data, social media data.
- ② **Stream Data Mining** : Extracting useful patterns from continuous, fast and infinite data streams in real-time. → **Example:** Online fraud detection, network traffic monitoring.
- ③ **Mining Unstructured Data** : Discovering patterns from unstructured data such as text, images, audio, video and documents. → **Example:** Sentiment analysis, image classification, document clustering.
- ④ **Text Mining & NLP** : Using Natural Language Processing to extract meaningful information from text data. → **Example:** Chatbots, opinion mining, automatic summarization.
- ⑤ **Social Media Mining** : Analyzing social media platforms to find trends, opinions, relationships and influence. → **Example:** Trend detection, brand monitoring, influencer identification.
- ⑥ **Graph Mining** : Discovering patterns in graph-structured data like social networks, web graphs and biological networks. → **Example:** Community detection, link prediction, recommendation systems.
- ⑦ **Deep Learning in Data Mining** : Using deep neural networks to automatically learn representations and discover complex patterns. → **Example:** Image recognition, speech recognition, fraud detection.
- ⑧ **Privacy-Preserving Data Mining** : Extracting patterns while preserving data privacy and security. → **Example:** Anonymization, differential privacy, secure multi-party computation.
- ⑨ **Cloud-Based Data Mining** : Performing data mining on cloud platforms for scalability, flexibility and cost efficiency. → **Example:** Mining large datasets using AWS, Google Cloud, Azure.
- ⑩ **Real-Time & Adaptive Mining** : Adapting to changes in data and environment to provide real-time and up-to-date patterns. → **Example:** Real-time recommendations, adaptive fraud detection.

In Short : Recent trends focus on big data, real-time analysis, unstructured data, deep learning, privacy and cloud computing to make data mining more powerful and useful.